ComplaintPulse: hearing every customer, in every language, before the problem goes viral

Challenge #2: AI-Powered Transaction Investigation and Resolution

ComplaintPulse listens to every complaint channel (call centre, app chat, social media, email, app reviews, branch forms) in English, Pidgin, Hausa, Yoruba and Igbo. It spots an emerging systemic problem hours before it peaks, names the likely root cause and the operational event behind it, and hands the owning team a ready-to-act brief, written by Nigeria's own language model, N-ATLaS, with a message to affected customers in their own language.

1. Problem & Opportunity

Nigerian banks count complaints; they don't hear them.

- Complaints escalated to the Central Bank keep rising: **23,129 in 2025, up from 20,925 in 2024**, and those are only the cases customers took to the regulator. CBN fined banks **₦1.69 billion in 2025**, including **₦430 million** specifically for ignoring customer complaints, and has forced back **₦115 billion** of failed-transaction money over ten years.
- The underlying causes are operational: failed transfers that are not reversed, switch and acquirer outages, ATM faults, USSD and OTP failures after app or telco changes. One fault produces thousands of complaints across channels, but each channel's team sees only its own slice.
- Banks monitor **ticket counts per day**. That rule fires a day late, raises many false alarms, and never says why.
- Customers who complain **in Hausa, Yoruba, Igbo or Pidgin, or by phone at a branch**, are the least visible in English-only analytics, and they are often the newly banked whose trust is easiest to lose.

The opportunity: treat every complaint, in every language, as a live sensor of the bank's operations, so the bank fixes the cause within hours and tells affected customers before they complain again.

2. Innovative Solution



Figure 1. ComplaintPulse pipeline: everything after the channels runs on one GPU.

1. **Privacy first.** Account numbers, BVN/NIN, phone numbers, card numbers and amounts are masked before any text is stored, embedded or shown to a model.
2. **Understanding every language with N-ATLaS.** Nigeria's national multilingual LLM (Llama-3 8B, trained on Hausa, Igbo, Yoruba and English) reads each complaint and returns structured JSON: issue, English summary, amount, urgency, language. It runs **on the bank's own GPU**; no complaint leaves the bank.

3. **Topics that ignore language.** Complaints are grouped by meaning with multilingual embeddings (BGE-M3), so new kinds of problems form new topics on their own.
4. **Three custom CUDA kernels:**
   - **K3 repost detection (MinHash + LSH):** a viral post copied 900 times counts once, not 900 times.
   - **K1 burst detection over a complaint cube:** topic × channel × state × customer segment, with every roll-up: 33,516 time series scored together every 15 minutes (seasonal baseline, Poisson surprise, CUSUM drift). It catches a spike in one state as well as a national one.
   - **K2 root-cause mining:** every attribute value (app version, device, payment switch, telco, ATM, card scheme, state, channel, segment, language) and **every pair of values** is scored by how much of the spike's *excess* it explains: "app 4.2.1 on Android explains 95% of the extra complaints, lift 25×".
5. **Event correlation:** the spike is matched to operational events (app releases, switch degradation, maintenance) that happened shortly before it.
6. **Systemic or not:** a spike without a concentrated cause (a salary-week volume rise, a random blip) is checked and **not escalated**. A concentrated one becomes an incident with an owner (app team, switch operations, ATM operations, card operations, billing, fraud).
7. **GenAI brief and customer message (N-ATLaS).** For each incident, N-ATLaS writes a brief for the owning team from the evidence only. A verifier checks **every number and complaint ID in the brief against the evidence**; errors are sent back to the model for correction, and a brief that still fails is held for analyst review instead of being sent. N-ATLaS also drafts a proactive message to affected customers **in Hausa, Yoruba, Igbo or English**; every message is approved by a person before it goes out.

3. Impact & Value (measured)

We built a 30-day synthetic stream of **249,553 complaints** modelled on a large Nigerian bank (diurnal and weekly patterns, salary-week surge, 37 states, 6 channels, 5 languages with code-switching), with **8 injected incidents** of known cause and timing, plus decoys: unrelated operational events, isolated high-value complaints and a viral repost storm. Real complaint data is confidential; the generator lets us measure against ground truth. To guard against tuning to one dataset, every result is also reported on a **held-out stream** (new random seed, incident times shifted by up to ±1.5 days, no retuning).

Result	ComplaintPulse	Daily-count rule (what banks use)
Incidents detected	**8 / 8** (held-out **8 / 8**)	8 / 8
Median time from onset to alert	**4.6 h** (held-out **5.8 h**)	18.6 h (held-out 22.6 h)
Alert before the incident peaks	**8 / 8** (held-out **8 / 8**), median 8.9 h early	2 / 8 (held-out 0 / 8)
False alarms	**0** (held-out **0**)	8 of 19 alerts (held-out 7 of 17)
Root cause ranked first	**8 / 8** (held-out **8 / 8**)	none given
Correct operational event linked	**8 / 8** (held-out **8 / 8**)	none given

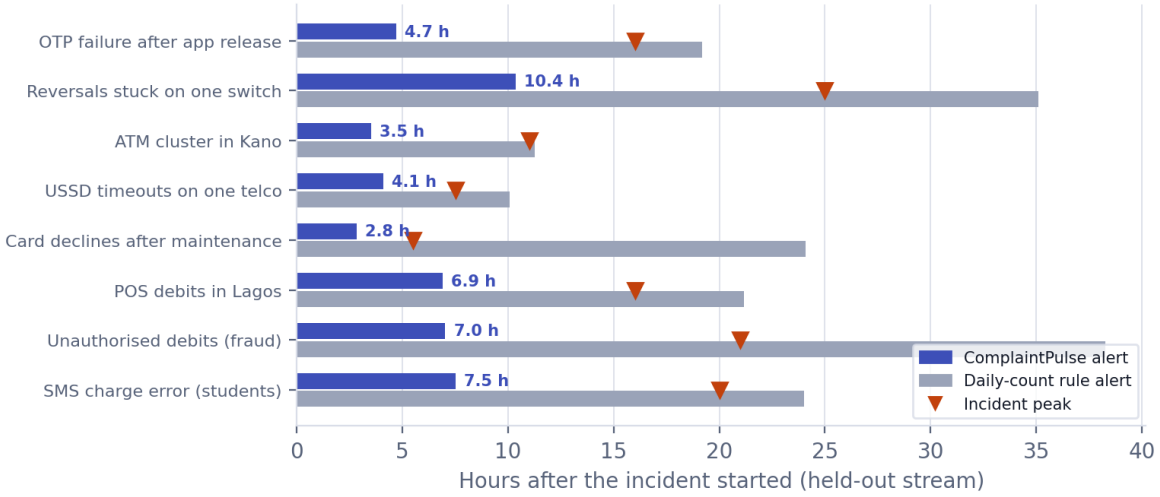


Figure 2. Time to alert on the held-out stream: ComplaintPulse vs the daily-count rule.

- **Repost storm:** K3 collapsed 895 of the 900 viral copies. Without it, the storm raises a false incident.
- **Salary-week surge and other blips:** 6 volume rises were checked and correctly not escalated.
- **Fully unsupervised topics** (no labels used anywhere; topics discovered from N-ATLaS's English summaries of each complaint): **8 of 8 incidents, 0 false alarms, root cause and event correct for all 8**, median 5.5 h to alert; the daily-count rule on the same topics raises 21 false alarms out of 33 alerts. Topics discovered from raw multilingual text also caught 8 of 8 on the held-out stream but missed the fraud campaign on the main stream, for the reason below.

GPU acceleration (A100 vs the same algorithms on CPU, outputs identical):

Kernel	Scale	CPU	GPU	Speed-up
K1 burst scoring	2.1 million series	51.7 ms	0.16 ms	**328×**
K2 root-cause mining	1.1 million attribute pairs	6.35 s	2.9 ms	**2,180×**
K3 MinHash	200,000 posts	1.27 s	6.7 ms	**189×**

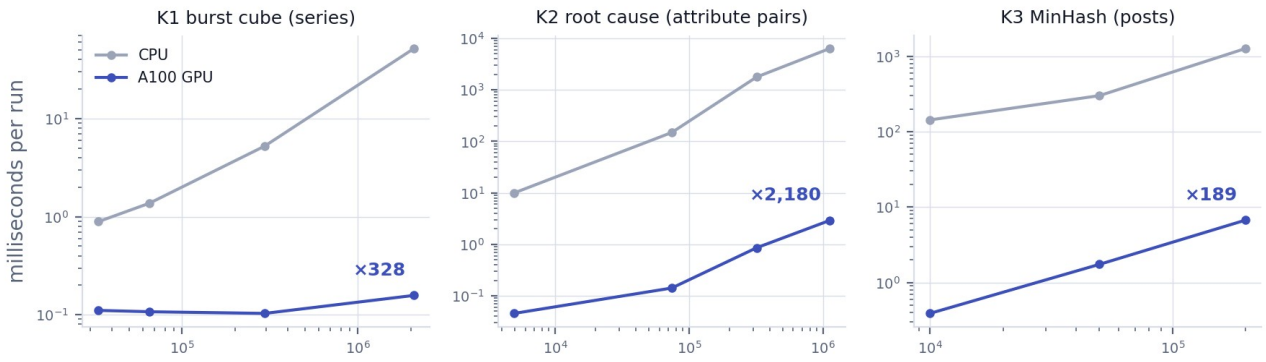


Figure 3. Same algorithms on CPU and A100 GPU; outputs identical.

K2 is what makes it possible to test every pair of attributes for every alert, across every topic, in real time.

Inclusion finding: N-ATLaS is what makes local-language customers count. Grouping complaints by their raw text, even with a strong multilingual embedding model, groups them by *language* as much as by *problem*: only 18–37% of Pidgin, Hausa, Yoruba and Igbo complaints joined the topic where English complaints about the same problem sat, so a local-language customer's complaint was split off

into a small topic of its own and could not raise an alarm. Having N-ATLaS first restate each complaint as a short English summary fixes this:

Language	Joins the right topic, raw text	Joins the right topic, after N-ATLaS
Pidgin	33%	**86%**
Hausa	37%	**77%**
Yoruba	18%	**78%**
Igbo	24%	**71%**
English	97%	91%

Topic purity rose from 0.89 to 0.96 (NMI 0.81 → 0.90), and the fraud campaign, whose complaints had been split by language, is now detected.

N-ATLaS understanding, per language (issue classified correctly, on the A100):

Language	Bank-style complaints	Complaints freely rewritten by N-ATLaS	All 249,553 complaints (macro-F1)
English	95%	83%	97% (97%)
Pidgin	92%	73%	89% (84%)
Hausa	88%	82%	82% (77%)
Yoruba	87%	73%	86% (77%)
Igbo	83%	63%	75% (68%)
All languages	89%	75%	91% (88%)

Sample: 300 complaints, 60 per language. Rewritten complaints are a harder, freer test (and a lower bound, since a rewrite can drift from the original meaning). N-ATLaS read all 27,511 distinct redacted complaints at 10.0 per second on one A100.

N-ATLaS briefs. 7 of the 8 incident briefs passed automatic verification of every number against the evidence. The 8th repeatedly wrote the money at stake ten times too small and was **held for analyst review**, which is exactly what the verifier is for: an 8-billion-parameter model will occasionally get a number wrong, and the bank must never send that number to an owner or a customer unchecked.

What we learned about local languages. N-ATLaS understands *what* customers complain about in every language (issue accuracy 75–97% across all 249,553 complaints, with zero unparseable outputs). Three limits are worth stating: it is not trained on **Nigerian Pidgin**, and its Pidgin customer messages were unusable, so Pidgin speakers receive simple English messages; it cannot reliably tell *which* language a complaint is in (Igbo and Pidgin were mostly labelled Yoruba), so the reply language should come from the customer's profile or a dedicated language-ID model, not from the LLM; and Yoruba and Igbo messages must be checked by native speakers, which the approval step provides.

4. Target Users & Implementation

Who benefits. Customer-experience and complaints teams, IT and operations incident teams, and compliance (CBN complaint reporting); then microfinance banks, payment service banks and fintechs. Customers benefit directly: problems are fixed hours earlier, refunds start sooner, and they hear from the bank in their own language before they need to complain again.

How it deploys.

- Reads exports or streams the bank already has: CRM tickets, contact-centre logs, email, social listening, app-store reviews, branch forms, plus the change and incident calendar.
- Runs on **one on-premises GPU server**, with open-weight models (N-ATLaS, BGE-M3). No customer data leaves the bank.
- Sends incidents to the existing service-management tool with an owner, a brief and a customer message for approval.

Rollout. (1) Back-test on 6–12 months of historical complaints and past incidents. (2) Live dashboard for one channel. (3) All channels with routed incidents and approved customer messages.

Feasibility today. The full pipeline runs end to end in our prototype on an A100: generator, redaction, N-ATLaS and BGE-M3 served on-premises with llama.cpp, the three CUDA kernels with CPU reference implementations, detection, evaluation, and a live console that replays the complaint stream and writes briefs on demand.

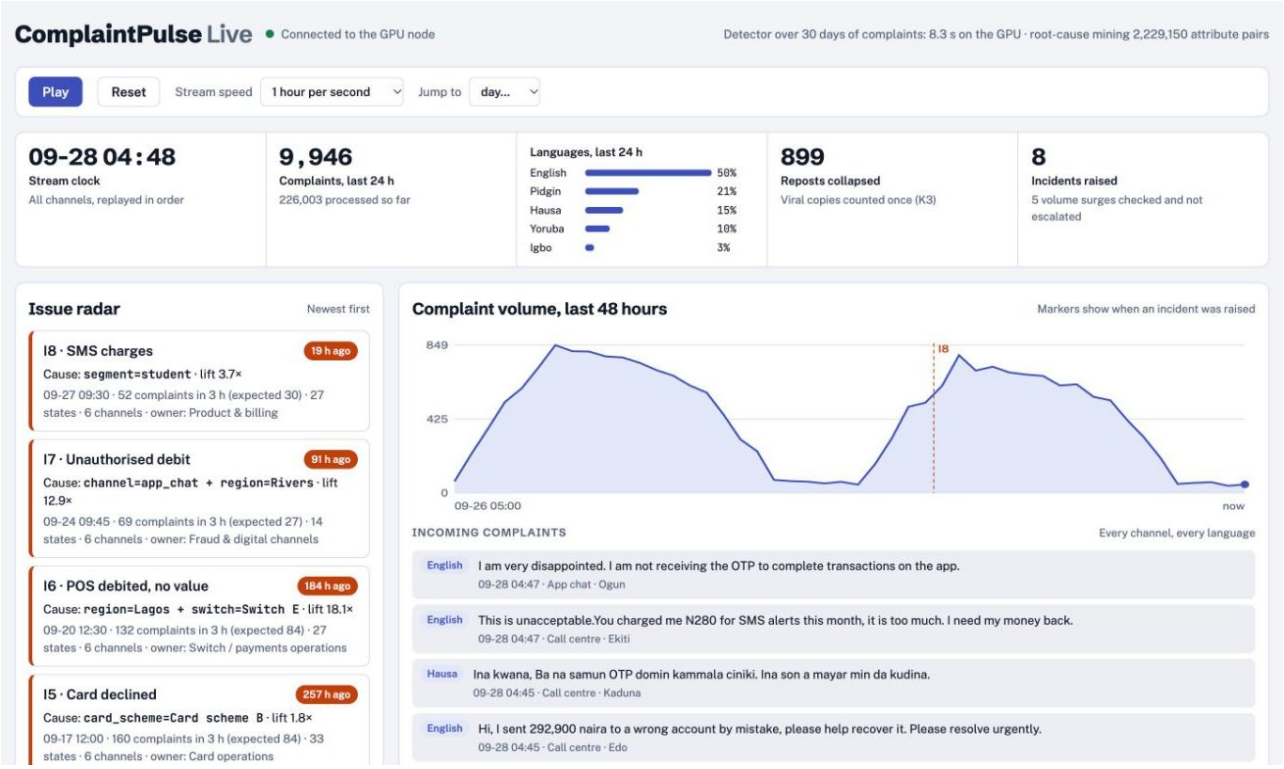


Figure 4. ComplaintPulse live console on the A100: issue radar, volume, multilingual feed.

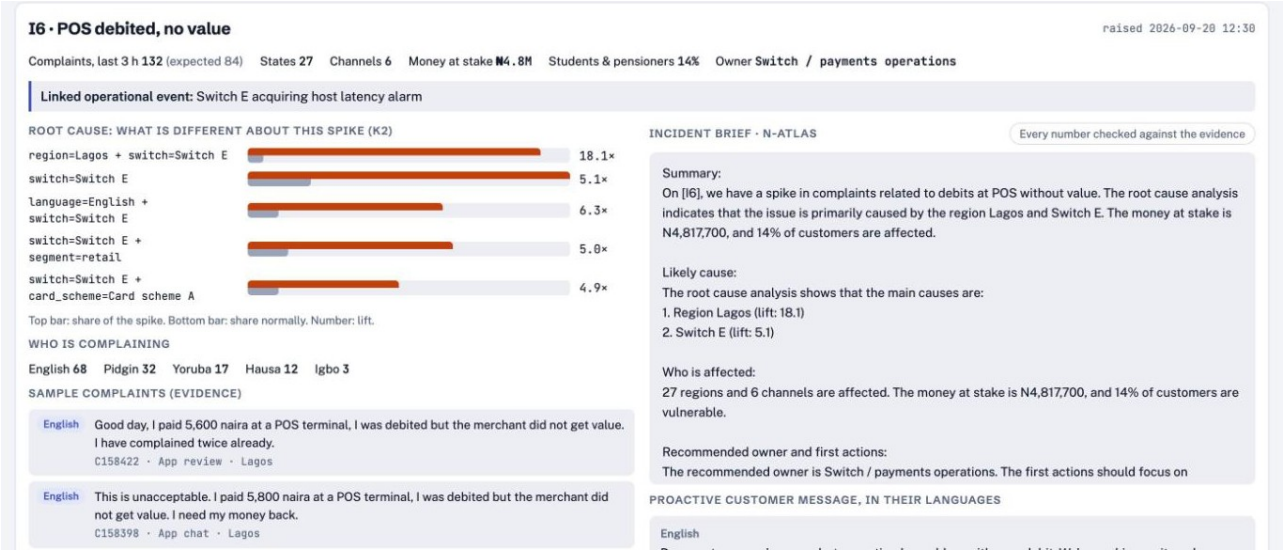


Figure 5. One incident: K2 root cause, linked operational event, routed owner, evidence, and the N-ATLaS brief that passed number verification.

Business model. Annual licence per institution tiered by complaint volume. N-ATLaS is free for up to 1,000 active users, which covers a bank's complaints and operations staff; larger deployments take a commercial licence from its publishers.

Link to MuleRadar. "Unauthorised debit" incidents (like the fraud campaign in our test stream) are passed to MuleRadar as seed accounts for mule-network analysis.

5. Team & Expertise

Member	Role	Contribution
Abdussamad Ali	CEO	Leads the venture and the AI and GPU build: detection kernels, evaluation, N-ATLaS integration; owns bank partnerships.
Abubakar Abdullahi	CTO	Architecture and deployment: ingestion from CRM, contact-centre and social feeds, on-premises GPU serving, security and PII redaction.
Umar Mohammed Bashir	Product Manager	Turns incidents into workflow: ownership routing, CBN complaint-handling requirements, customer-message approval, pilot plan.
Abdulbasit Idris Ali	UI Designer	Designs the console: issue radar, root-cause view, multilingual complaint feed, brief and message review.

Sources

- CBN complaints and fines 2024–2025: Prime Business Africa, Nairametrics
- Failed-transaction refunds over 10 years: Blueprint
- N-ATLaS model card and terms: NCAIR1/N-ATLaS